

Expt 6

Finite Word length Effects

6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB

Objectives:

1. To represent a digital signal using a finite word length format in MATLAB.
2. To perform truncation and rounding operations on the finite word length representation of the signal.
3. To evaluate and compare the errors resulting from truncation and rounding techniques.
4. To investigate the impact of finite precision arithmetic on the accuracy of digital signal representation.

Procedure:

1. Open MATLAB and create a new script file.
2. Define the sampling frequency, signal frequency, and time vector.
3. Generate a discrete-time sinusoidal signal using the MATLAB `sin()` function.
4. Select an appropriate finite word length (e.g., 8 bits) for representing the signal samples.
5. Scale the signal according to the selected word length to obtain a finite-precision representation.
6. Apply the truncation operation using the `fix()` function to remove the fractional part of the scaled signal samples.
7. Apply the rounding operation using the `round()` function to round the scaled signal samples to the nearest integer value.
8. Convert the truncated and rounded samples back to their original amplitude range by dividing them by the scaling factor.
9. Compute the truncation error by subtracting the truncated signal from the original signal.
10. Compute the rounding error by subtracting the rounded signal from the original signal.
11. Calculate the Mean Square Error (MSE) and maximum error for both truncation and rounding operations.
12. Compare the error values obtained from truncation and rounding.
13. Analyze the effect of finite word length on signal representation and evaluate which

method provides better accuracy.

Program (Matlab code):

```
clc;
```

```
clear;
```

```
close all;
```

```
n = 0:79;
```

```
x = sin(2*pi*n/40);
```

```
q = 8;
```

```
%% Quantization
```

```
xt = floor(x*q)/q;    % Truncation
```

```
xr = round(x*q)/q;    % Rounding
```

```
%% Quantization Error
```

```
et = x - xt;          % Truncation error
```

```
er = x - xr;          % Rounding error
```

```
%% Fig 1 - Original vs Quantized
```

```
figure;
```

```
plot(n, x, 'b', n, xr, 'r', 'LineWidth', 1.5);
```

```
grid on;
```

```
title('Original vs Quantized');
```

```
legend('Original', 'Quantized');
```

```
%% Fig 2 - Truncation vs Rounding
```

```
figure;
```

```
plot(n, x, 'b', n, xt, 'r', n, xr, 'g', 'LineWidth', 1.5);
```

```
grid on;
```

```
title('Truncation vs Rounding');
```

```
legend('Original', 'Truncated', 'Rounded');
```

```
%% Fig 3 - Quantization Error
```

```
figure;
```

```
stem(n, et, 'r');
```

```
hold on;
```

```
stem(n, er, 'g');
```

```
grid on;
```

```
title('Quantization Error');
```

```
legend('Truncation', 'Rounding');
```

```
%% Fig 4 - MSE Comparison
```

```
figure;
```

```
bar([mean(et.^2), mean(er.^2)]);
```

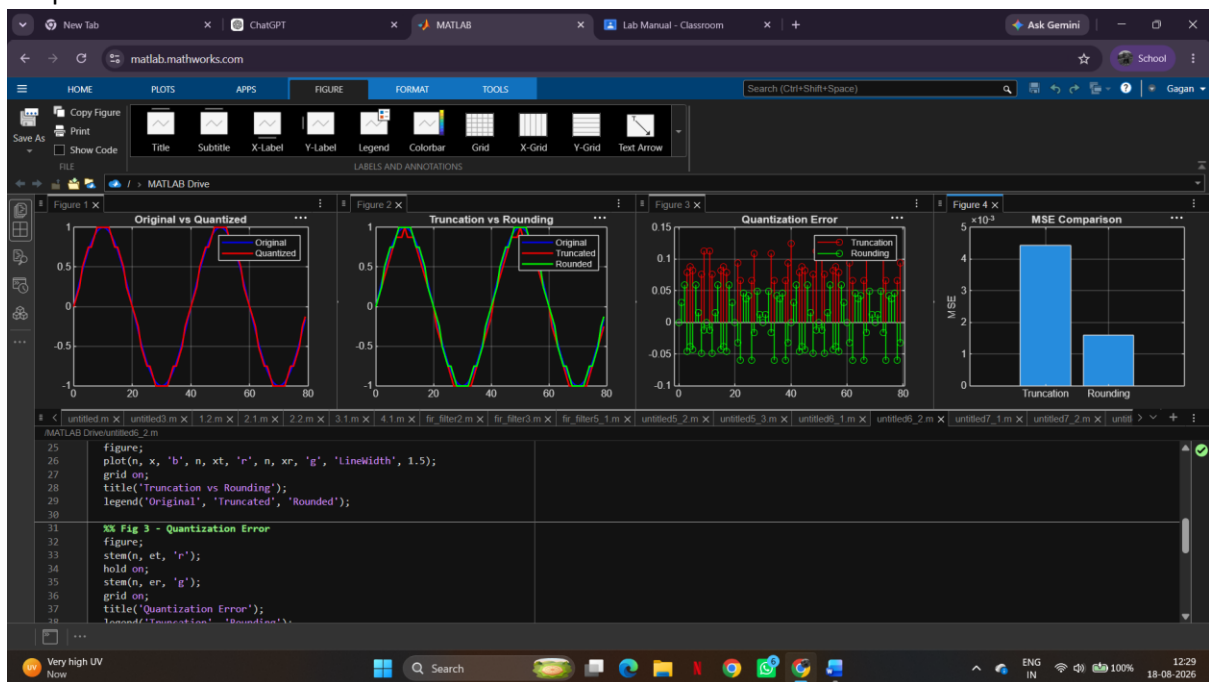
```
set(gca, 'XTickLabel', {'Truncation', 'Rounding'});
```

```
grid on;
```

```
title('MSE Comparison');
```

```
ylabel('MSE');
```

Output:



Result:

1. The digital signal was successfully represented using a finite word length format in MATLAB.
2. Truncation and rounding operations were implemented on the signal samples.
3. The errors introduced by truncation and rounding were computed and compared, showing that rounding generally produces smaller errors than truncation.
4. The effect of finite precision on signal representation was analyzed, demonstrating that reduced word length increases representation error and may affect signal accuracy.